INCH-POUND

MIL-C-24643/33C
14 March 1994
SUPERSEDING
MIL-C-24643/33B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWU (INCLUDING VARIATION LS2SWUA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2SWU (unarmored), LS2SWUA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch) tin coated. Maximum diameter 0.050 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair. 1/
Fifth - Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
* Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

1/ For LS2SWU-1. omit seventh, eighth and ninth.

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Construction (watertight) (continued)

Eighth - The required number of pairs (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS2SWU

* Tenth - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LS2SWUA

* Tenth - Cross-linked polyolefin jacket. (see table I for thickness).

* Eleventh - Braided metal armor

* TABLE I. Details.

Military 1/ part no. M24643/33	Size 2/	Number of pairs	Diameter over completed pair nominal (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
					TYPE LS2SWU		TYPE LS2SWUA	
					min	max	min	max
-01UN	LS2SWU-1	1	0.213	0.015	0.240	0.255	0.290	0.305
-02UN	LS2SWU-3	3	.213	.075	.670	.710	.720	.760
-03UN	LS2SWU-7	7	.213	.075	.860	.910	.910	.960
-04UN	LS2SWU-12	12	.213	.088	1.130	1.200	1.190	1.250
-05UN	LS2SWU-19	19	.213	.100	1.292	1.380	1.342	1.420
-06UN	LS2SWU-24	24	.213	.100	1.500	1.590	1.550	1.640
-07UN	LS2SWU-30	30	.213	.100	1.670	1.760	1.720	1.810
-08UN	LS2SWU-37	37	.213	.125	1.785	1.870	1.825	1.920
-09UN	LS2SWU-61	61	.213	.125	2.205	2.300	2.255	2.250

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS2SWU and LS2SWUA.

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EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.47
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield 3/	500
* Shield to water	500
* Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
Component shield to shield	100
Conductor and shield continuity	No failure

3/ Not required for type LS2SWU-1.

Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum	1.0
Capacitance	
Mutual capacitance - 1 megahertz (MHz), pF/ft, maximum	30
Capacitance unbalance - percent maximum	8
Characteristic impedance - at 1MHz, ohms	75 ± 5
Cable attenuation - power loss (at sinusoidal frequency of 1 MHz), decibels (db)/ 100 FT, maximum	1.5

Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80

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EXAMINATION AND TESTS (continued):	Requirements
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Group C: (continued)	Requirements
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket - (types LS2SWU only) cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LS2SWU only) - conformance to material, construction and coverage	No failure
* Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

Size	Feet (nominal)
1 through 12	1500
19 through 24	1000
30 through 61	500

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